SupplyMind — PRD (v1 MVP)

What it is

SupplyMind is an AI-powered inventory risk platform. It predicts stock-out risk for a business's products and explains *why* that risk exists in plain language, instead of just handing over a probability score.

Who it's for (v1)

Small and mid-sized businesses that sell physical goods and manage their own inventory and vendors (retailers, distributors). v1 is a standalone, self-serve web product — no existing ERP or inventory system required.

How it works, end to end

1. **Sign up & onboard.** Business creates an account, enters basic company info.
2. **Inventory setup.** They add products (SKU, stock level, reorder threshold) and link each to a vendor (name, typical lead time, backup vendor if any).
3. **Risk factor selection.** The business picks which disruption factors are relevant to them from a list — e.g. vendor reliability, Naira exchange rate, demand seasonality. No forced, irrelevant factors.
4. **Risk model built per business**, using their products, vendors, and chosen risk factors.
5. **Ongoing monitoring.** The system tracks sales data and the chosen external signals, recalculates stock-out risk continuously, and raises an alert with a plain-language explanation when risk crosses a threshold.
6. **Action.** Small routine reorders are flagged as suggestions (auto-trigger later, once trusted). Bigger risks are shown to the business owner with the explanation, for a manual call.

v1 risk factors (buildable, verifiable)

- **Vendor reliability** — tracked from the business's own order history (how often a vendor delivers late).
- **Naira exchange rate** — free, reliable public APIs.

- **Demand seasonality** — known high-sales periods, holidays.

Factors like port congestion are out of scope for v1: no reliable, low-cost, real-time data source exists yet. Revisit later if a good data provider is found.

Explanations, not just scores

Every risk alert comes with: the top contributing factors, their relative weight, and a recommended action in plain language — not just a bare percentage.

Two paths to market

- **Standalone (primary focus for v1):** business signs up, builds their inventory in our dashboard, SupplyMind monitors it.
- **API integration (later / secondary):** business with an existing system gets an API key, feeds in their data, dashboard setup becomes optional. Not a v1 priority.

AI approach

- **Risk engine (we build it):** Python backend that calculates stock-out risk and the factors behind it, using a causal graph (pgmpy/dowhy) plus a simple demand forecaster. Every number on the dashboard comes from here.
- **Explanation layer (we rent it):** an external LLM API (e.g. OpenAI) turns the engine's verified numbers into plain-language explanations and recommendations. It never calculates or invents figures, and its output is validated against structured fields.
- **Cost control:** call the LLM only when a risk changes or a user asks; cache explanations; track usage per business.
- **No own language model** in v1.

Goals and non-goals

Goal: a business can sign up, enter its inventory, and get trustworthy stock-out alerts with a plain-language reason and a suggested action, without owning any other software.

Not in v1:

- Automated purchasing (alerts and suggestions only)
- API/ERP integration (later)

- Port congestion and other hard-to-source external signals
- Chat assistant and free-text scenario questions (later, once alerts are trusted)
- Training our own language model

v1 requirements

#	Feature	What it must do
1	Accounts	Sign up, log in, one workspace per business
2	Onboarding wizard	Company info, then products, then vendors, then risk factors
3	Inventory	Add/edit products (SKU, stock, reorder threshold) and vendors (lead time, backup vendor); CSV import for sales history
4	Order tracking	Log orders and actual delivery dates so vendor reliability is measured from real data
5	External signals	Naira exchange rate pulled automatically; seasonality calendar
6	Risk engine	Per-product stock-out risk and top contributing factors, recalculated on a schedule and on any data change
7	Explanations	Plain-language reason and suggested action per alert, validated against the engine's numbers
8	Dashboard	Risk list sorted by urgency, product detail view, alert history
9	Notifications	Email alert when a product crosses its risk threshold
10	Usage and cost tracking	LLM calls counted per business, with a cap

Build order

1. **Foundation:** accounts, database, inventory and vendor screens, CSV import.

2. **Baseline risk:** stock divided by daily sales, compared with vendor lead time. Alerts on the dashboard and by email.
3. **Real signals:** order tracking, vendor reliability score, Naira rate, seasonality.
4. **Causal layer:** causal graph over those factors, with the top-factor breakdown behind each alert.
5. **Explanations:** LLM wording layer with structured, validated output.
6. **Pilot:** 3 to 5 real businesses, then fix what breaks.

Definition of done

v1 is finished when every box below is ticked. The numeric thresholds are proposed; the team should confirm or change them.

- [ ] A new business can sign up and reach a working dashboard in under 20 minutes, unaided
- [ ] All 10 requirements above work end to end in production
- [ ] Risk numbers match a hand calculation on a test set of at least 20 products
- [ ] Every alert shows its top contributing factors and a suggested action
- [ ] No explanation shows a figure that differs from the engine's output (checked automatically)
- [ ] Naira rate and vendor reliability update without manual work
- [ ] Backtest: on past data, alerts fire before stock-outs more often than a plain "stock divided by sales" baseline
- [ ] 3 to 5 pilot businesses use it for 4 weeks, and most say the alerts were useful
- [ ] LLM cost per business per month is measured and within the agreed cap
- [ ] Known issues from the pilot are fixed or logged

Risks

Risk	Mitigation
Small businesses have little sales history	CSV import; use vendor and lead-time data first; show lower confidence when history is thin
One wrong alert costs a user money	Alerts and suggestions only; show confidence; no auto-ordering

Risk	Mitigation
LLM states a wrong figure	Numbers come only from the backend; validate structured output
Causal graph is only as good as its inputs	Start with three verifiable factors; show data freshness on each
LLM cost grows with users	Cache, call only on change, per-business cap

Open questions for the team

- How do we source and verify vendor reliability data cleanly from day one?
- What's the minimum viable dashboard — what screens does a business truly need at launch?
- Pricing model: subscription vs. usage-based, and where the AI API cost sits in that.
- How do we build trust before enabling auto-triggered reorders?